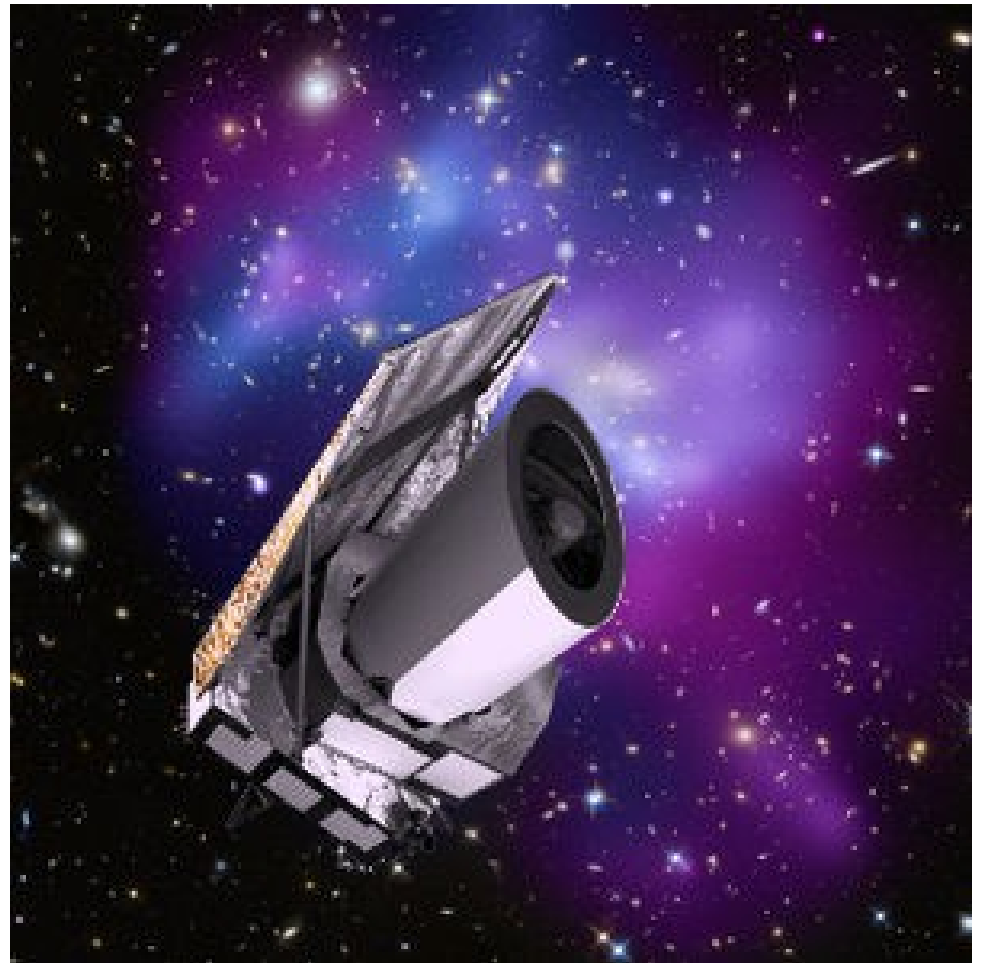


Charge Transfer Inefficiency (CTI) Calibration In Euclid

James Nightingale
Richard Massey

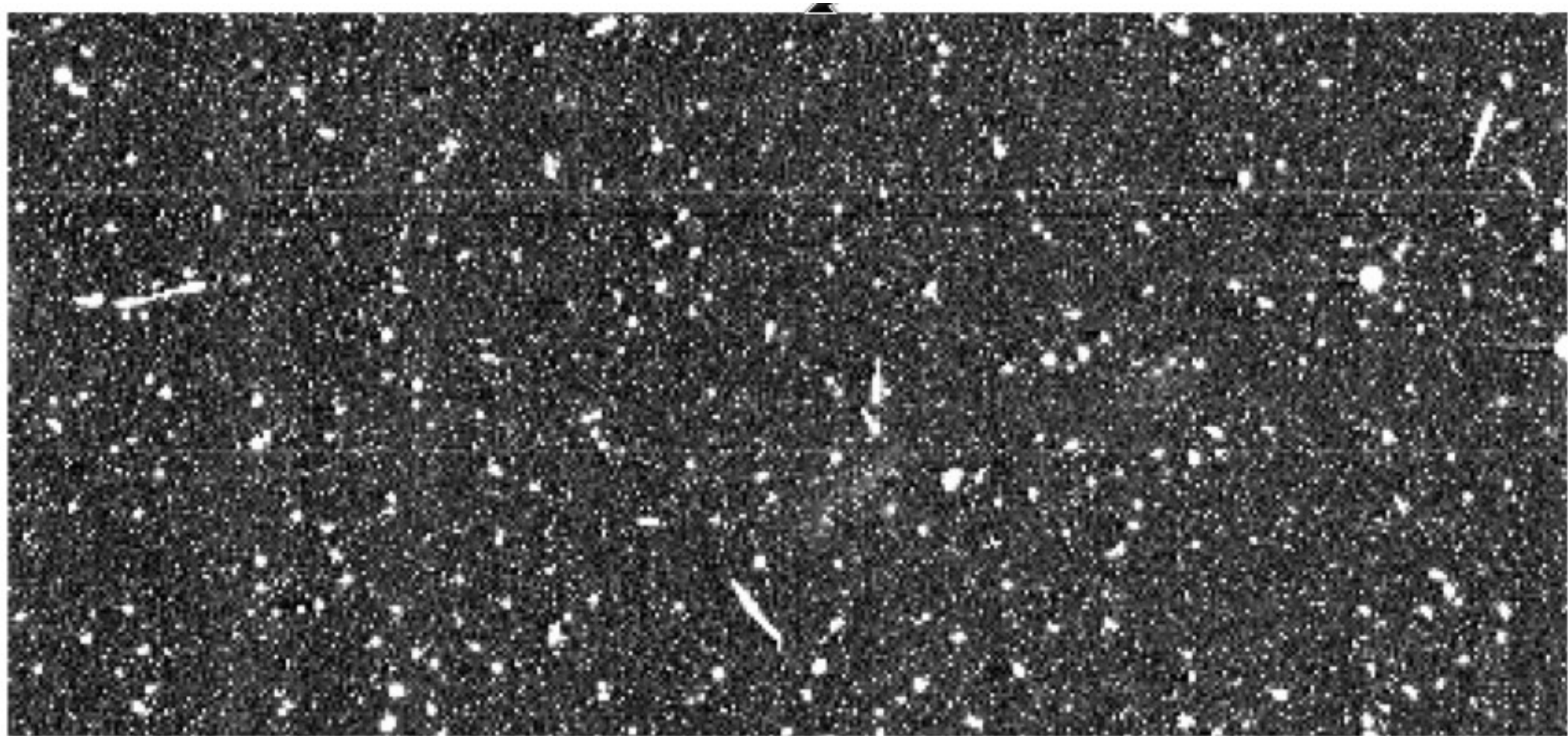
OU-VIS
OU-SHE



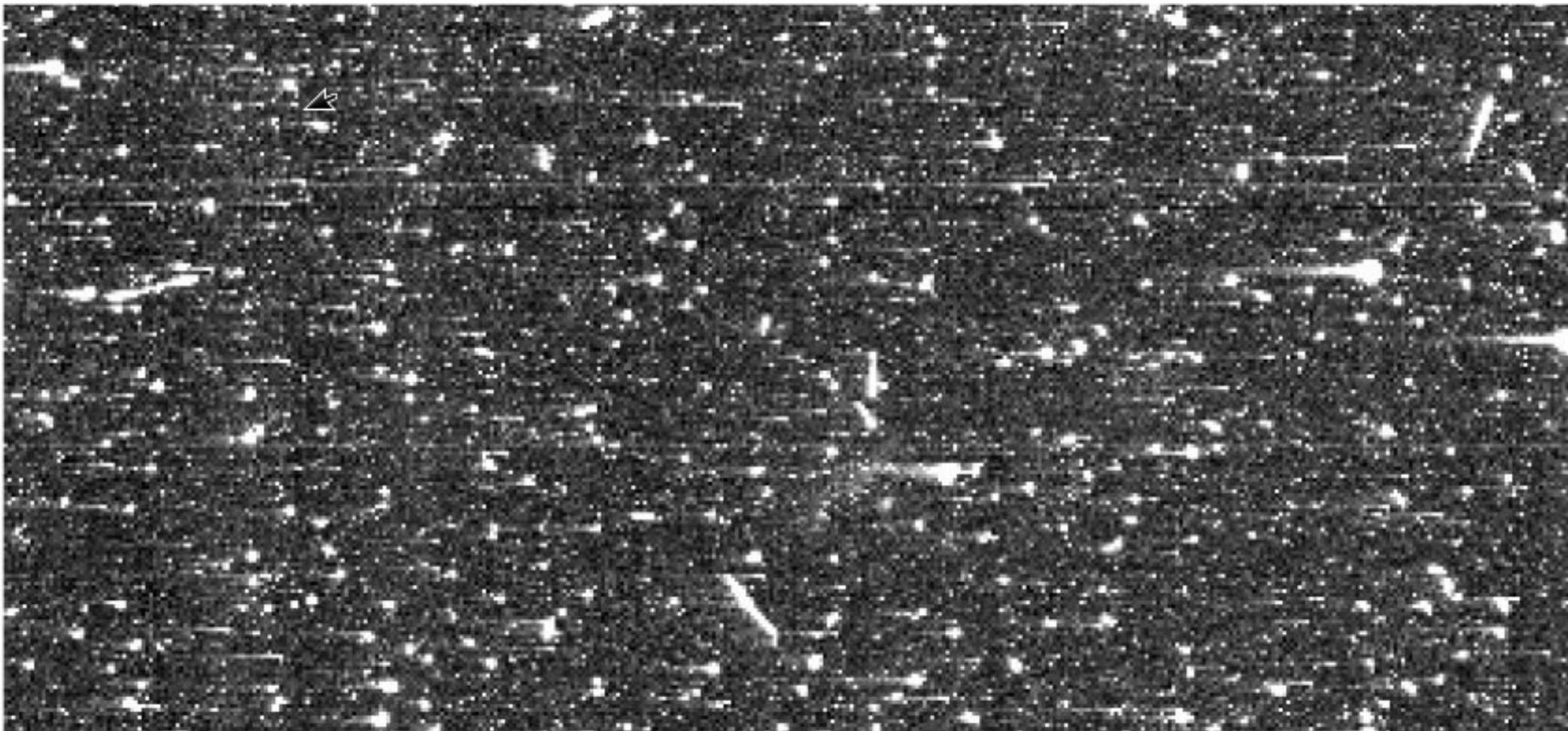
What is CTI?

- **Harsh radiation / high energy particles constantly bombard Euclid's CCD.**
- **These damage its silicon lattice, forming 'traps', in the silicon lattice which hold photo-electrons for a short time period.**
- **If this time is of order (or longer) than the CCD clocking speed, the captured electron will be released in a different pixel during read-out.**

Original Exposure



CTI Trailed Exposure



How do we correct CTI?

- Use **arCTIc** (Massey 2011 / 2014), tried and tested on HST imaging.
- **Input** – raw image data before most other data processing steps (includes CTI trails).
- **Process** – model pixel-by-pixel read-out of CCD, including CTI trailing due to traps (properties of which are defined by CTI model).
- **Output** – expected trails due to CTI (use a forward modeling approach to correct CTI).

How does arCTIc model CTI?

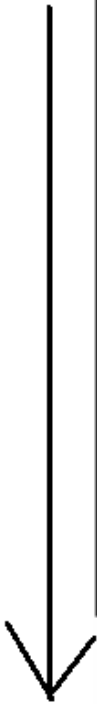
- **Physical equations / functions:**
 - Describe the capture & release of electrons, etc.
 - Well understood from solid state physics / lab.
- **Model describing trap properties on each CCD:**
 - Exponential release time constant τ_r of each trap.
 - Density ρ_t of each trap species.
 - The volume fill parameter β .
- **With 3 trap species for both parallel and serial directions this gives us 14+ parameters to calibrate for!**

How do we calibrate the CTI model parameters?

- **Input an image that we know what it looks like.**
 - The image we observe (after read-out) will include CTI trails.
 - However, we know the true input image, so can calibrate the CTI model by fitting the observed image.

'Input' Charge Injection Lines

Parallel Direction



Serial Direction



'Observed' Charge Injection Lines

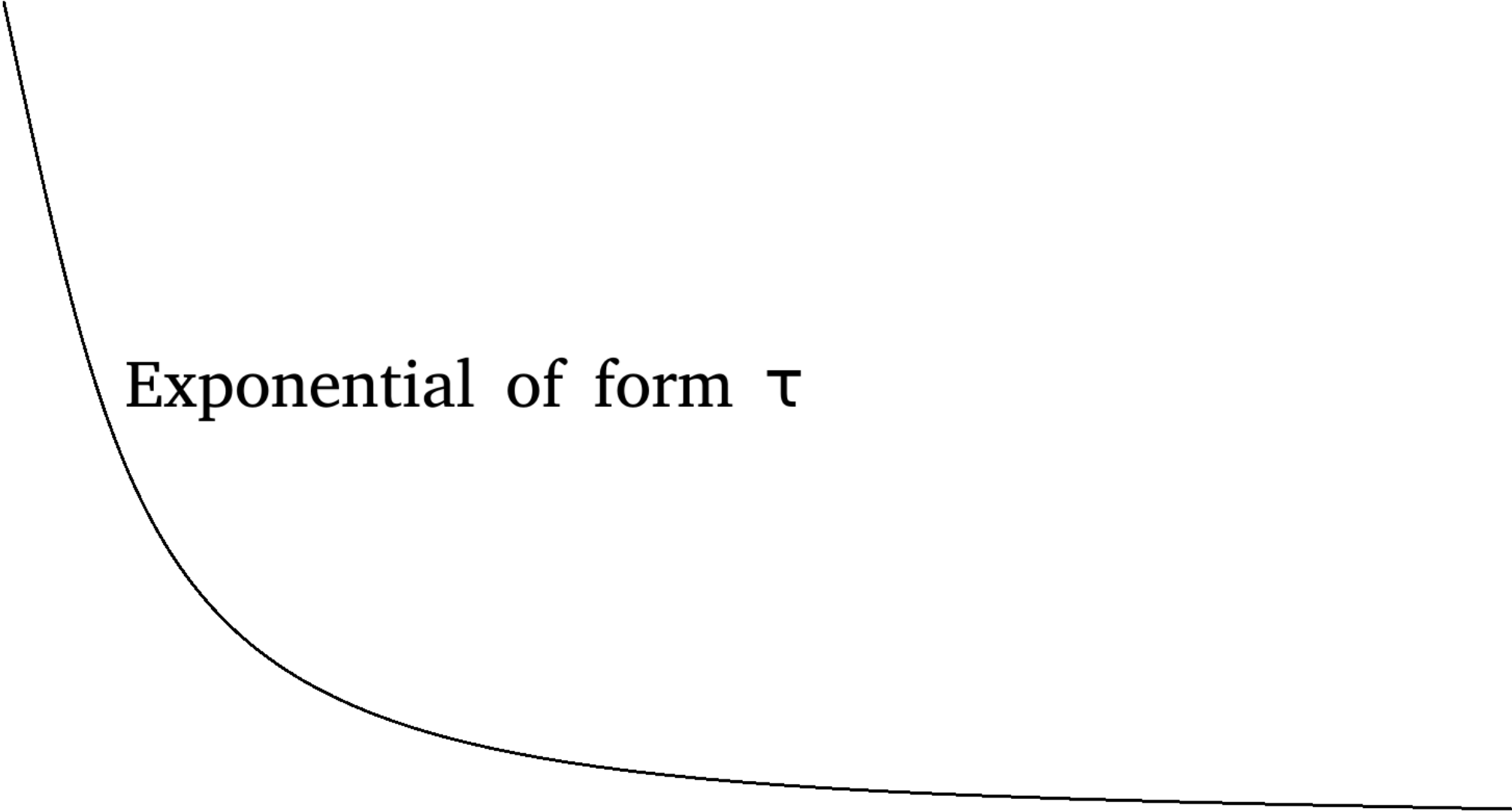
Parallel Direction



Serial Direction



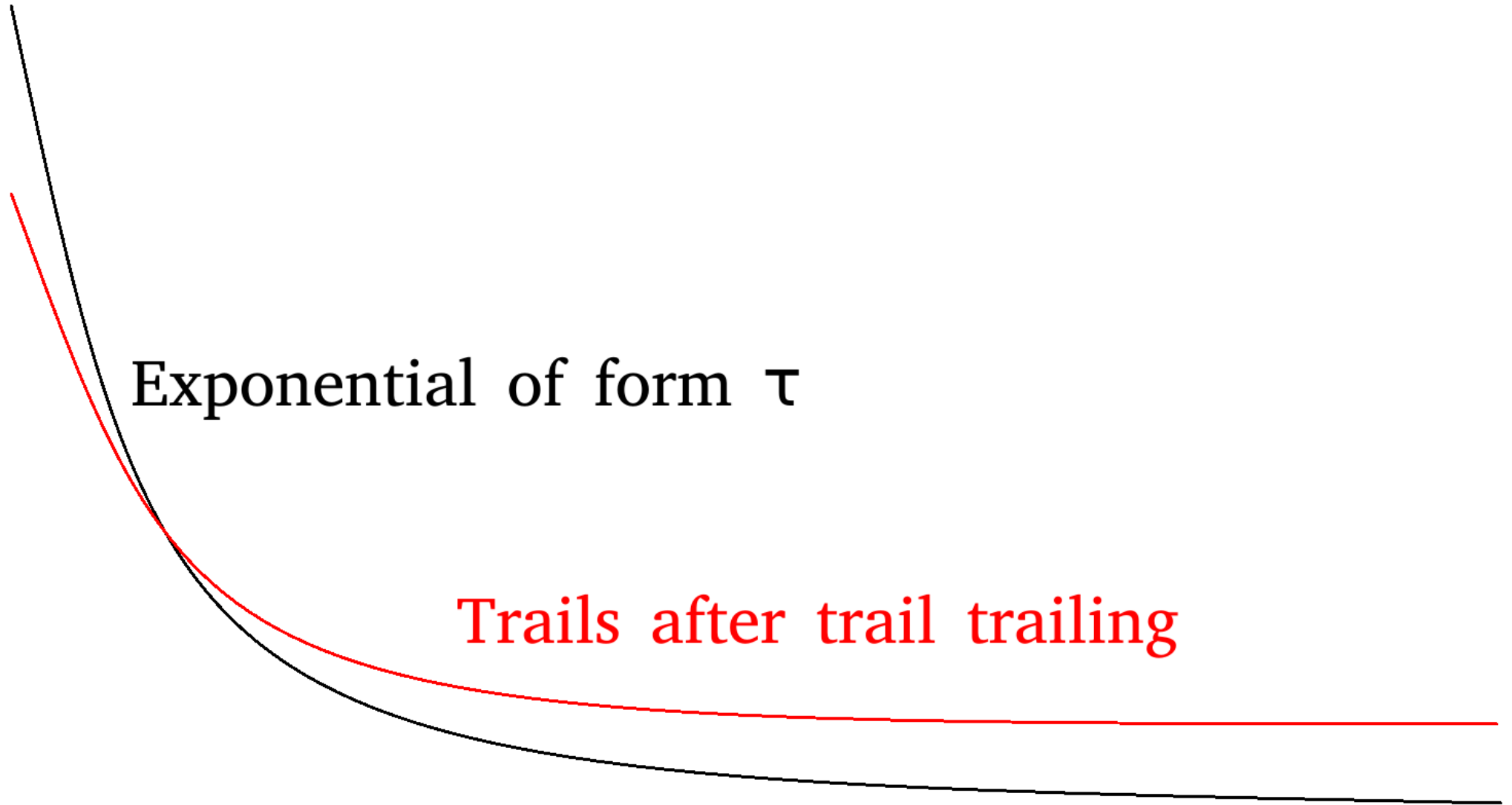
Issue - Trailing of trails



A graph illustrating an exponential decay curve. The curve starts at a high value on the left and decreases rapidly, approaching the horizontal axis as it moves to the right. The text "Exponential of form τ " is placed near the curve.

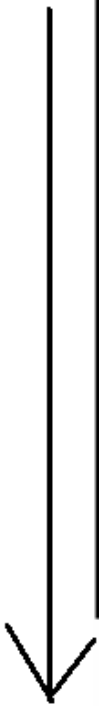
Exponential of form τ

Issue - Trailing of trails



'Model' Charge Injection Lines

Parallel Direction



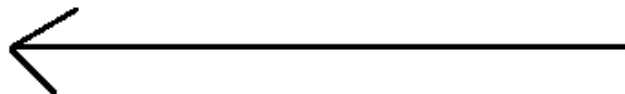
Serial Direction

Run Through arCTlc (Incorrect Model)

Parallel Direction

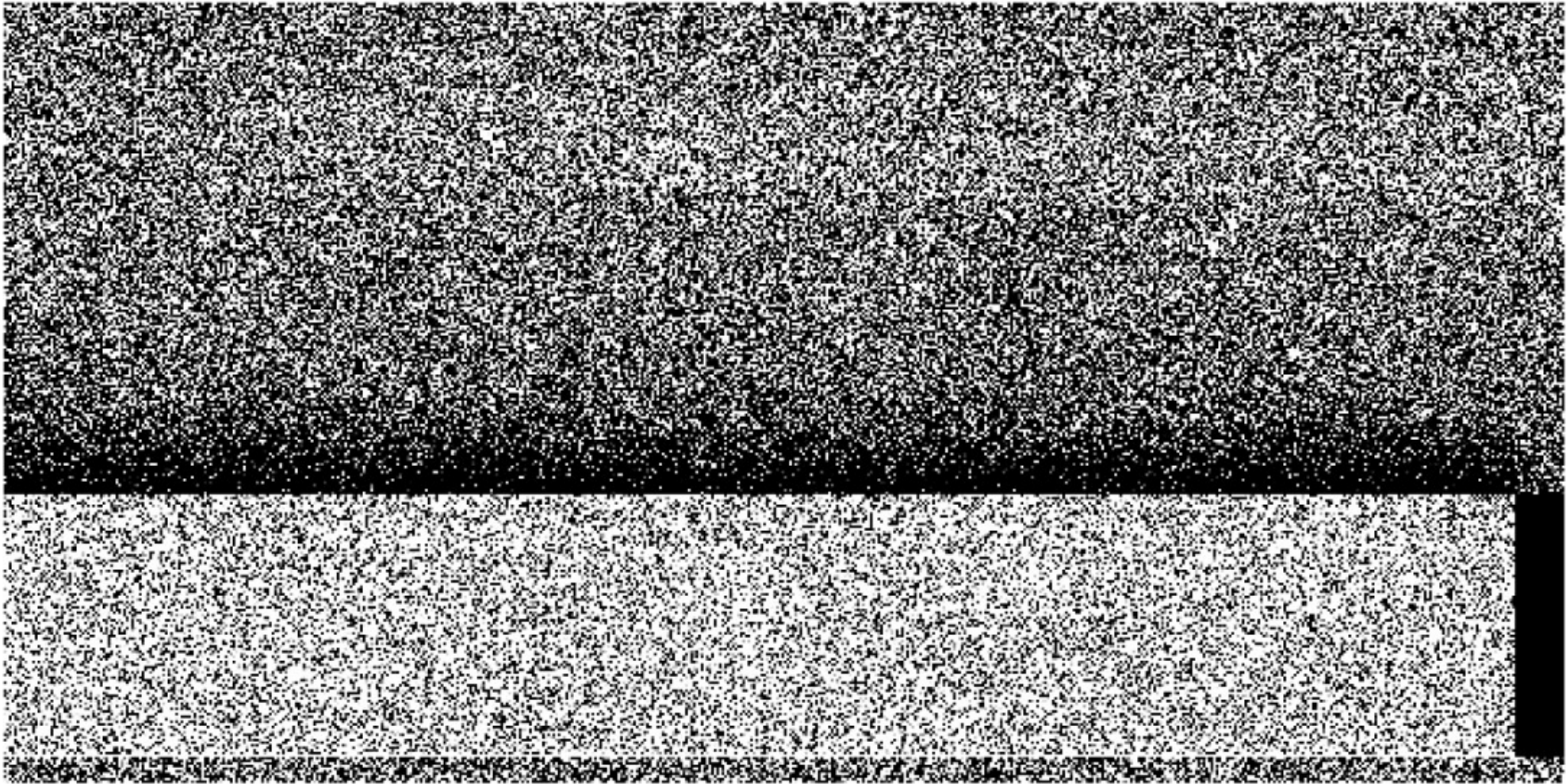


Serial Direction



Compute Residuals (incorrect model)

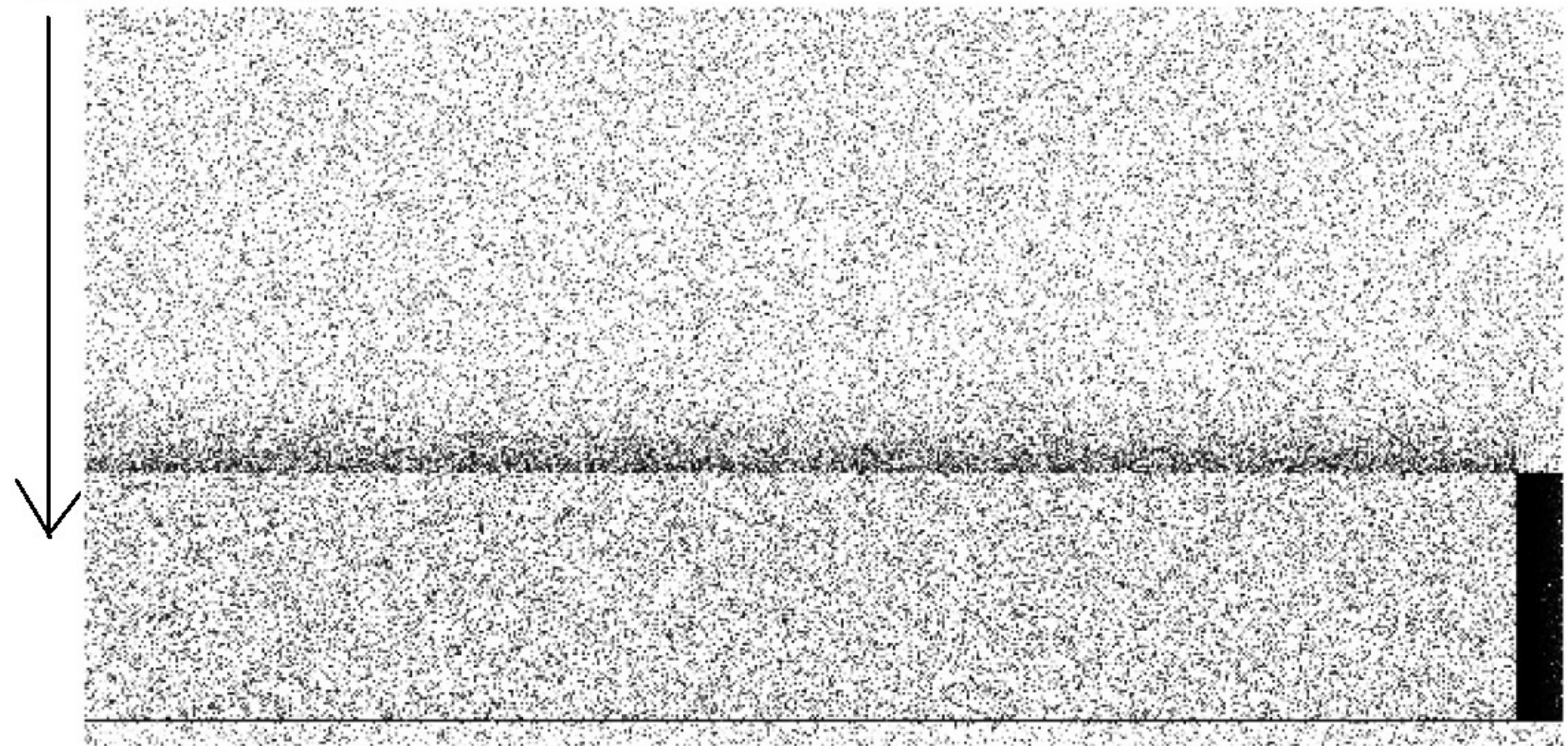
Parallel Direction



Serial Direction

Compute χ^2 (incorrect model)

Parallel Direction



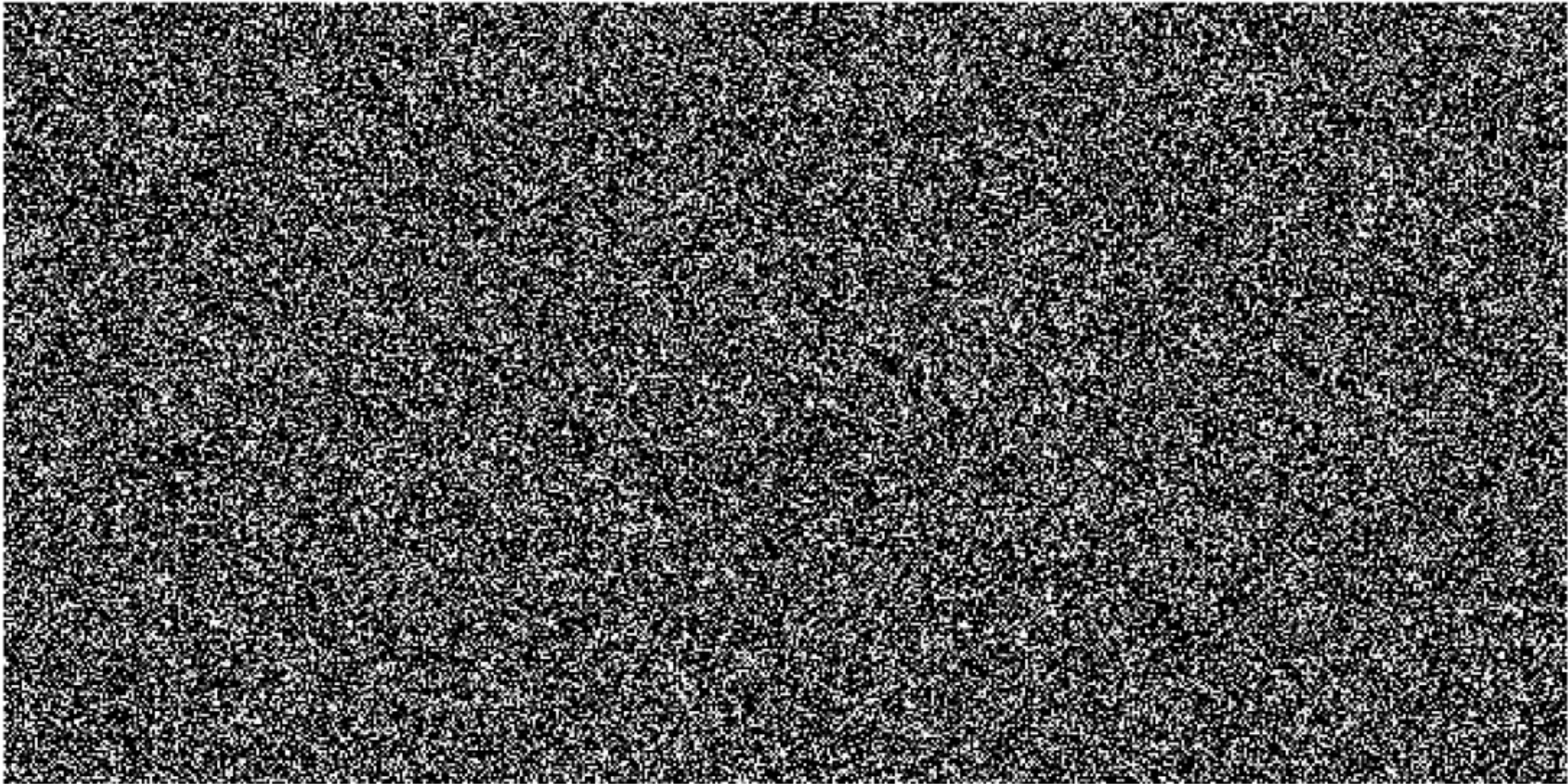
Serial Direction

Non-linear Search - MultiNest

- **Calibration is thus a standard non-linear optimization problem.**
 - Complex and degenerate parameter space necessitates advanced non-linear search.
 - Use nested sampling approach with algorithm MultiNest.

Residuals (After calibration)

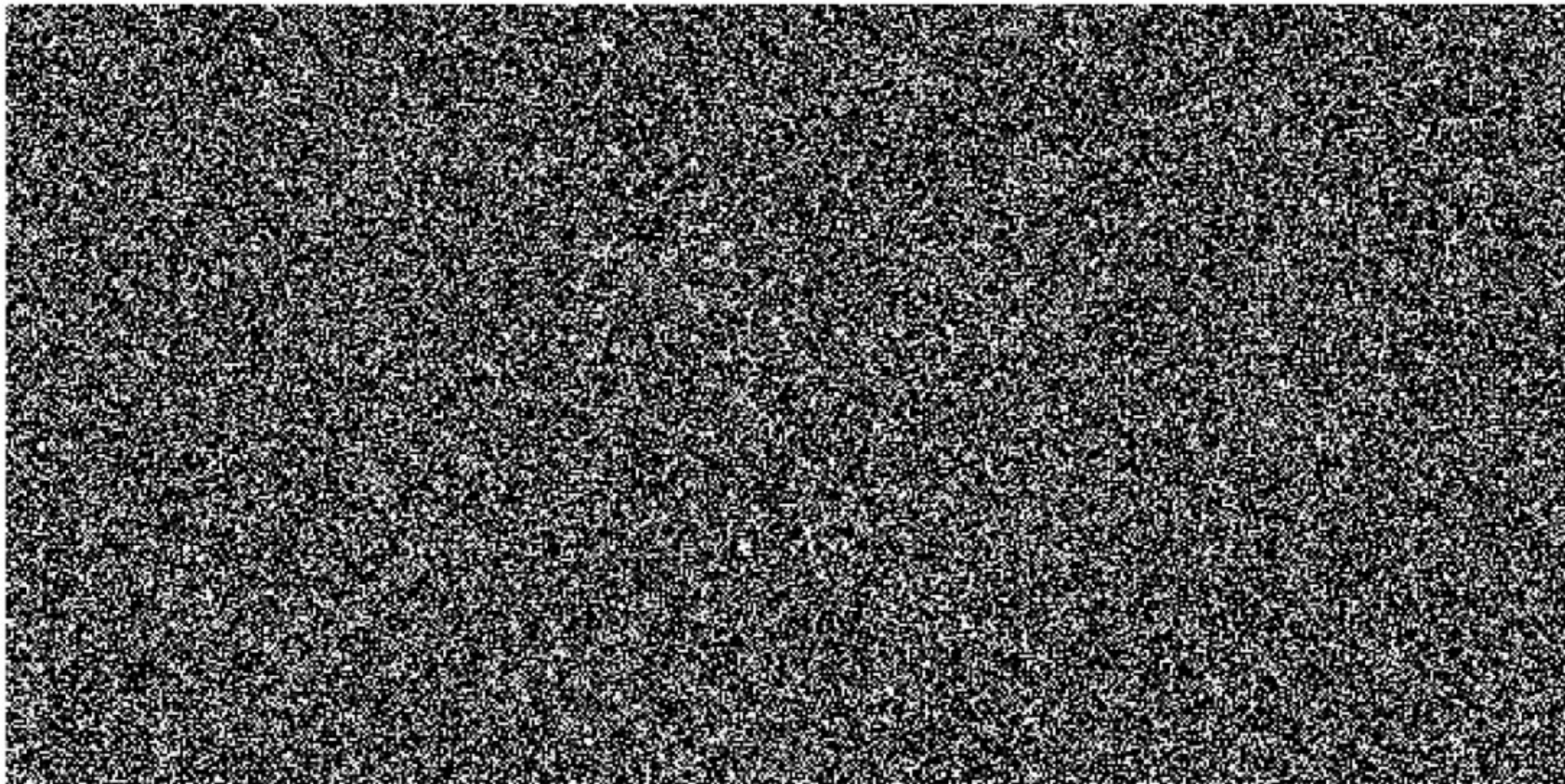
Parallel Direction



Serial Direction

χ^2 (After calibration)

Parallel Direction



Serial Direction

Multiple Charge Injection Images

- **Illustration showed one charge injection image.**
 - Calibration in fact takes an array of charge injection images, each with a different normalization of electrons (100, 250, 500, 1000, etc., up to the full well capacity).
 - Inputs all through arCTIc and uses all to compute χ^2 .
- **This is required to calibrate the β parameters**
- **Method already does this – results I showed used 6 charge injection images!**

CTI Calibration

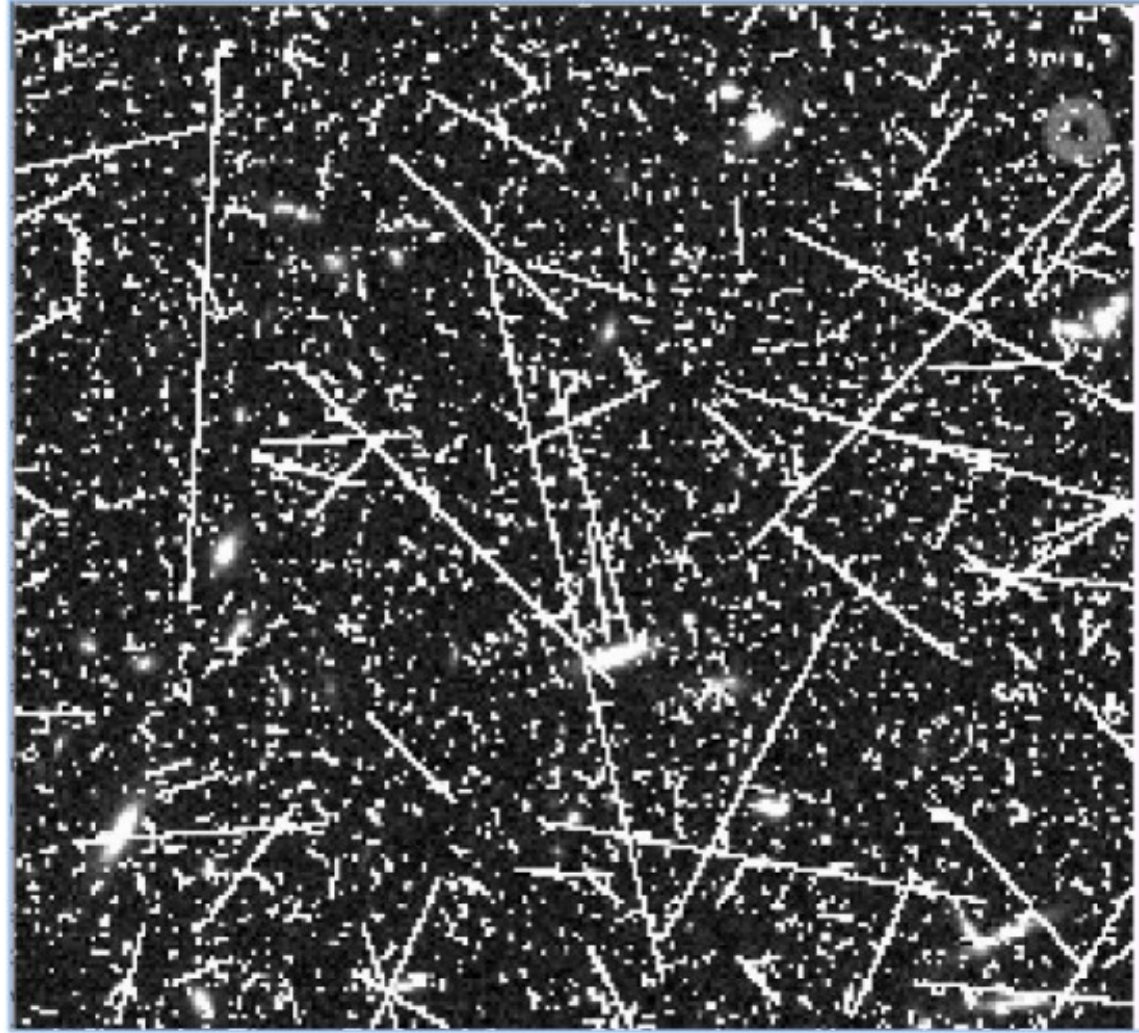
- Working successfully!
- Capturing 'trail of trails' effect.

Additional CTI Information

- **Science Imaging** – will use parallel / serial overscan of science imaging for calibrating lowest charge levels (i.e. background sky).
- **Non-uniformity** – Requirements on the uniformity of charge injection lines. Provided they are (relatively) smooth can model and include in CTI calibration.
- **Parallel + Serial CTI** – At a low level, CTI in these two directions ‘mixes’. Method models both simultaneously, likely invoke a hybrid approach in the future.
- **Calibration Database** – Will ultimately create a database of CTI model parameters based on calibrations taken every day.

Cosmic Rays

- Will be present in charge injection imaging.
- Impossible to include in model image.
- Will next explore best approach to handle them.
 - Can we ignore them?
 - Masking / flagging?



CTI calibration precision / accuracy

- **Can we calibrate the CTI model parameters at the accuracy / precision to meet requirements? (Israel et. al 2015).**
- **This is the main question I aim to answer in the next 6 months.**
- **If we can't, what can we do about it?**

Questions?

Extra Slides

Science Image Calibration

- **May be unable to inject charge at very low levels (e.g. background sky, < 100 electrons).**
 - Therefore can't calibrate β at this level via charge injection imaging.
 - This is where the majority of galaxies for shear measurement will be observed.
- **Therefore, will also use science imaging to calibrate CTI, fitting trails in the parallel / serial overscan from the background sky.**

Simultaneous Parallel / Serial Calibration

Parallel Direction

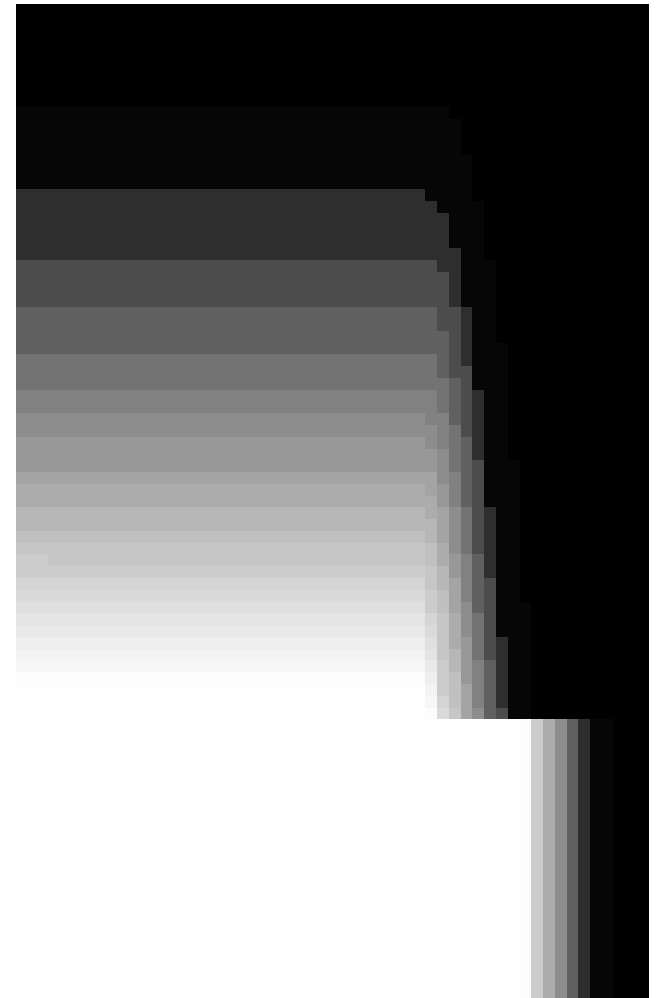


Serial Direction



Simultaneous Parallel / Serial Calibration

- **Parallel and serial CTI mix with one another.**
- **Most evident at corner of charge injection lines.**
- **Happens over the whole image at a very low level.**
- **Most likely move towards a hybrid approach, fitting each separate first before together.**



Non-uniformity

- **Charge injection lines won't be perfectly uniform.**
- **Should be able to detect non-uniformity in the charge injection data and incorporate into the model image given to arctic.**
- **Potential to lead to errors, but a benefit is it will sample a wider range of normalizations giving better constraints on β .**